

HapticPalmrest: Haptic Feedback through the Palm for the Laptop Keyboard

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ABSTRACT

Programmable haptic feedback on touchscreen keyboards enriches user experiences but is hard to realize for physical keyboards because this requires individually augmenting each key with an actuator. As an alternative approach, we propose HapticPalmrest, where haptic feedback for a physical keyboard is provided to the palms. This is particularly feasible for a laptop environment, where users usually rest their palms while interacting with the keyboard. To verify the feasibility of the approach, we conducted two user studies. The first study showed that at least one palm was on palmrest for more than 90% of key interaction time. The second study showed a vibration power of 1.17 g (peak-to-peak) and a duration of 4 ms was sufficient for reliable perception of palmrest vibrations during keyboard interaction. We finally demonstrated the potential of such an approach by designing Dwell+ Key, an application that extends the function of each key by enabling timed dwelling operations.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

KEYWORDS

haptic feedback, laptop keyboard, sensory substitution

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1 INTRODUCTION

A physical keyboard provides inherent tactile feedback, but the tactile feedback is unvarying. On the other hand, a touchscreen keyboard lacks inherent tactile feedback but, augmented with haptic technology, can provide varying, programmable tactile feedback. For example, Park et al. [10] designed 72 tactile stimuli for virtual button feedback on a mobile touch screen. Kim and Lee [3] designed tactile stimuli to reproduce the tactile feeling of different mechanical buttons. The applications of such programmable tactile feedback envisioned by these studies may be equally valuable for a physical

keyboard. A few past studies presented a physical keyboard with programmable tactile feedback. For example, Hoffmann et al. [2] augmented each key in a physical keyboard with a solenoid so that it can have a high resistance when the user attempts to press a wrong key, thereby reducing typing errors. Wu and Smith [13] augmented keys in a thin keypad with programmed piezoelectric vibrations to improve the tactile quality of the thin keypad. Ogawa et al. [7] presented an extended physical button that could deliver a variety of sensations with a programmable vibration. Park et al. [9] further explored the design space of such augmented buttons while changing vibration frequency, amplitude, duration, and envelope. Park et al. [8] presented a text-entry application scenario in VR where an augmented physical button was used to cue a typing error. Each key was augmented individually with an active haptic element in these studies. As a physical keyboard usually contains 87 to 105 keys, embedding an actuator in every key may not be practical when the hardware's required mechanical and electronic complexity is considered.

Alternatively, we considered providing tactile feedback to the palm, instead of the keys, in response to key presses. This may be particularly feasible in the laptop environment, where users usually rest their lower palms on the palmrest areas below the keyboard. The palm area has been utilized to deliver tactile information in prior studies. Minamizawa et al. [6] provided tactile stimuli on the palm to augment bimanual manipulation in VR environments. Guzererler et al. [1] applied skin stretch tactile feedback on the palm. Son and Park [11] proposed a wearable haptic interface that provided tactile feedback at both fingertips and palms. These studies suggest the effectiveness of providing tactile feedback on the palm. However, no previous studies investigated tactile feedback on the palm to enhance physical keyboard interaction on a laptop.

In this paper, we propose a concept of HapticPalmrest, providing tactile feedback on the palm area to augment keyboard interaction on a laptop and the results of two user studies that we conducted to verify the feasibility of the concept. In the first user study, we observed when and where users stayed their palms on the palmrest areas while performing everyday tasks on the laptop. We then conducted another user study to inform the design of effective vibrotactile stimuli for HapticPalmrest. We measured detection rates for different vibration parameters and target key positions. In addition, we designed and implemented the Dwell+ Key technique as an application of novel keyboard input augmented with haptic cues of HapticPalmrest.

Our contribution is three-fold: (1) empirical findings on users' palm-resting behaviors during laptop keyboard interaction, (2) empirical findings on users' tactile perception of vibrotactile stimuli through palmrest areas, and (3) haptically augmented key input

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(a) Laptop with touch-sensing plates (User Study 1) (b) Laptop with vibrating plates (User Study 2)

Figure 1: Apparatus for User Studies 1 and 2

technique on laptop environment that shows a possible application of HapticPalmrest.

2 USER STUDY 1: PALM-RESTING BEHAVIORS

The study aimed to quantitatively understand users' palm-resting behaviors during laptop keyboard interaction.

2.1 Apparatus

We made two touch-sensing plates with 3x4 electrodes and attached them to the two palmrest areas, as shown in Figure 1a. Each plate uses a 12-key capacitive touch sensor (MPR121, Freescale) and outputs 3x4 analog values, ranging from 0 to 100, which are proportional to the touch areas on the 3x4 electrodes in the plate.

2.2 Tasks

We set the following three criteria to select tasks that participants would perform: (1) The tasks should be commonly experienced by laptop users. (2) Either keyboard or mouse interaction should occur continuously. (3) Mouse and keyboard usage patterns should differ across the selected tasks. Based on the criteria, we chose the following three tasks: (1) Find products in given categories that satisfy a given price range and add them to the shopping cart. (2) Create presentation slides explaining one product selected in the previous task. (3) Summarize two given documents.

2.3 Participants

We recruited six participants (2 males, 4 females, average age = 21.33) from our university. All were right-handed and touch typers. Each was paid 15,000 KRW for participation.

2.4 Procedure

Participants read and signed an informed consent form. They were seated and instructed to perform the three tasks described in section 2.2 using a provided laptop. We required participants to use the laptop touchpad instead of an external mouse for mouse interaction. While they perform the three tasks, the laptop recorded the touch-sensing plate outputs. The whole experiment took approximately 60 minutes for each participant.

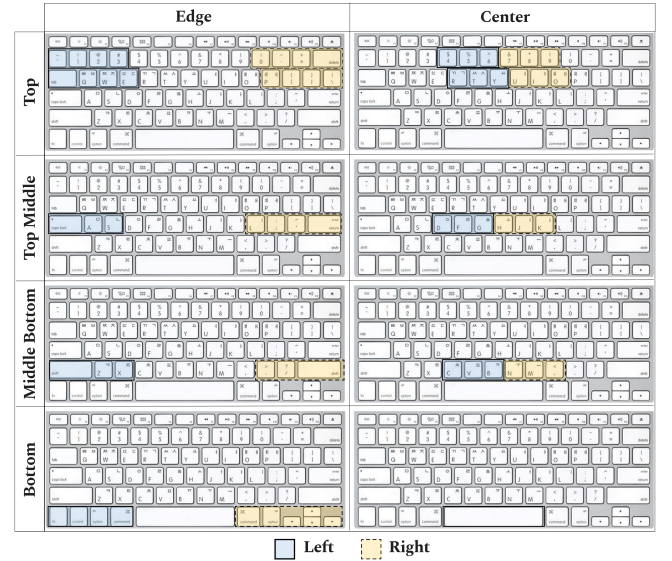


Figure 2: 4x2 key groups. The left and right subgroups are marked in blue (solid border) and yellow (dash border), respectively.

2.5 Result

We estimated palm-resting areas and palm-resting durations from the logged sensor outputs. We defined the following metrics.

RestingArea: the average of the outputs of all 12 touch sensor outputs for a touch-sensing plate (0 – 100%). This is calculated for each plate.

LeftHandDuration (*RightHandDuration*): the percentage of the time for which the left (right) *RestingAreas* was over 5%.

LeftOrRightDuration: the percentage of the time for which the left or right *RestingAreas* was over 5%.

Based on an in-lab preliminary test, we chose the threshold 5% as the minimum palm-resting area required for a user to perceive the vibration of the palmrest.

We estimated these metrics for different key groups that were expected to require different hand postures. For this purpose, we classified keys according to their horizontal positions (edge or center) and vertical positions (top, top middle, middle bottom, and bottom). As a result, we could define 2x4 key groups, as shown in Figure 2. Each group is further divided into left and right subgroups, as indicated by the color of the keys in the figure, according to the hand generally used to access the keys. The bottom center group contains the space key only, and the space key is not colored because it can be accessed by either hand.

2.5.1 RestingAreas. Figure 3 shows the heatmap visualizations of the left and right touch-sensing plate outputs (3x4 cells for each) and the computed *RestingArea*'s for the eight key groups. *RestingArea*'s appear to depend on the horizontal and vertical key positions. *RestingArea*'s for the edge key groups appear smaller than that of the center groups. This may be because, as the participants pressed keys in the central area, the palm moved to the inner part of the palmrest. In addition, for the keys in the upper



Figure 3: Heatmaps that show the average outputs of the touch-sensing plates. The averages of the 12 heatmap cell values (*RestingArea's*) are shown below. The left (right) heatmaps show the outputs of the left (right) touch-sensing plates when the left (right) hand typed the keys in the left (right) key subgroups.

Key Groups	Top		Top Middle		Middle Bottom		Bottom	
	Edge	Center	Edge	Center	Edge	Center	Edge	Center
Left	97.5	97	97.4	97.1	93.7	94.5	80.2	97.8
Right	93.1	99.4	97.1	98.2	100	92.8	91.2	

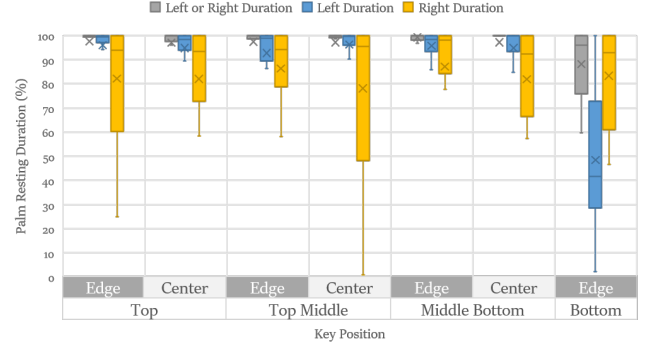
Table 1: *LeftOrRightDurations* for the left and right subgroups of the eight key groups.

part of the keyboard, the touch area moved upward, and the overall amount of resting area increased. For the space key, participants usually press it with their thumb, and therefore *RestingArea's* for the bottom center group were comparable to that of the top groups.

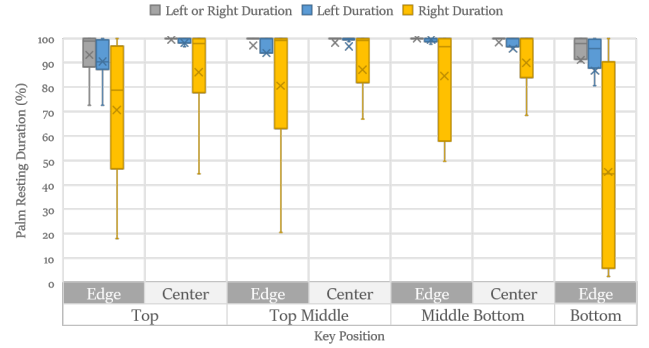
2.5.2 RestingDuration. As shown in Table 1, *LeftOrRightDuration* was more than 90% of the time for all key groups, except for the left subgroup of the bottom edge group (80.2%). As shown in Figure 4, the *RightHandDuration* for the right subgroup of the bottom edge group was short (mean 37.7%), but *LeftHandDuration* was long (mean 86.8%). Therefore, the *LeftOrRightDuration* was high (mean 91.2%) for such group.

2.6 Conclusion

The study's main result was that *RestingDuration* was longer than 90% for most keys. The worst case was for the modifier keys for which *RestingArea* and *RestingDuration* were both low. This suggests that laptop users would be able to perceive haptic feedback provided at the palmrest for most of the keys. Another main result was that *RestingArea* depended on key positions. This suggests



(a) Palm Resting Duration for the left subgroups



(b) Palm Resting Duration for the right subgroups

Figure 4: Box Plots showing *LeftOrRightDuration*, *LeftDuration*, and *RightDuration* for the (a) left and (b) right subgroups of the seven key groups (excluding the bottom center group).

that the perception of tactile feedback on the palm would depend on the key positions.

3 USER STUDY 2: PALMREST VIBRATION PERCEPTION

The study aimed to investigate how well users could perceive the vibration of the palmrest for different vibrations and key positions. The result of this study is intended to inform the design of effective vibrotactile stimuli for HapticPalmRest.

3.1 Apparatus

Figure 1b shows the apparatus used for this user study. We attached two acrylic plates to the left and right palmrest areas using adhesive sponge tape. The sizes of the left and right plates were 104 mm X 80 mm and 130 mm X 80mm, respectively, which fit the size of the respective palmrest areas of the laptop. Each plate was actuated by a voicecoil actuator (Taptic Engine for iPhone 6). Each actuator was driven by a 250 Hz square wave signal. The vibration frequency was chosen because the human fingertips are known to be most sensitive at this frequency [12]. The amplitude of the driving signal was varied according to the experiment condition, as described below.

3.2 Study Design

We conducted a within-subjects study. The independent variables were two vibration parameters (duration (*Duration*) and strength (*Strength*)) and target key positions (*Position*). The dependent variable was the percentage of participants to perceive a vibration (*Detection*).

Duration had three levels (4 ms, 8 ms, and 12 ms), and *Strength* also had three levels (10%, 30%, and 50% of the maximum strength). The three duration levels correspond to 1, 2, and 3 cycles of 250 Hz vibration, respectively. As the sizes of the left and right plates differed, the required electrical strengths for the left and right actuators had to be different to create the same vibration strength. After a calibration process, we measured the average peak-to-peak values of the acceleration along the vibration axis of the actuator in the centers of the two vibrating plates. They were (3.407, 3.606), (2.203, 1.993), (1.291, 1.054), and (0.31, 0.269) for 100%, 50%, 30%, and 10% strength levels, respectively. The tuples are for the left and right plates, and the unit of the values is g , which is 9.8 m/s^2 .

Position had seven levels corresponding to the eight key groups in User Study 1, excluding the bottom center group. We excluded the bottom center group because the *RestingArea* was comparable to that of the top groups.

3.3 Tasks

We used three tasks: both-hand task, left-hand task, and right-hand task. In the both-hand task, the participant was instructed to press the key given on the screen and release it when they perceived a vibration on the palm provided by the system immediately in response to the key press. If the participant could not sense vibration and did not release the key in one second, the trial was declared a failure, and the following key to the press was shown. The key to press was chosen randomly from the seven key groups. The left-hand (right-hand) task was the same as the both-hand task, except that the participant had to perform the task using the left (right) hand with the other hand removed from the palmrest. In addition, keys to the press were selected from the left (right) subgroup of the seven key groups.

When participants type using both hands, they may have a higher chance of perceiving a vibration than a single hand. Therefore, the one-hand tasks represent worse cases than the both-hand task for HapticPalmrest. The both-hand task represents the actual keyboard usage more than the one-hand task. However, our focus in this user study was on the latter because our goal was to verify the feasibility of HapticPalmrest.

3.4 Participants

We recruited 15 participants (12 males, 3 females, average age = 22.47) from our university. Each was paid 10,000 KRW for participation.

3.5 Procedure

The participants read and signed the informed consent form. They then conducted five sessions. In each session, they performed a set of trials with the both-hand task, another set with the left-hand task, and the third set with the right-hand task. In each set of trials, the participants performed a trial for each of the 3x3 vibration

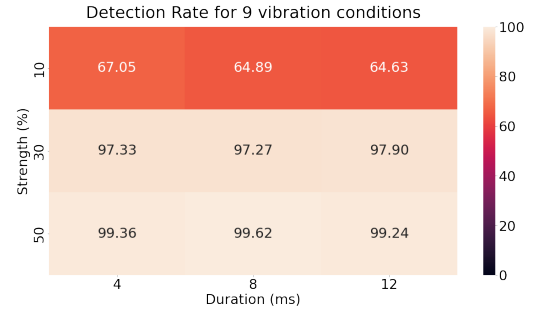


Figure 5: Heatmap of *Detection* for 3×3 vibration conditions

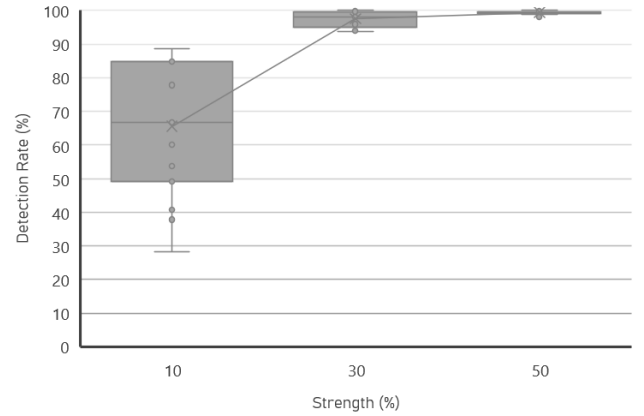


Figure 6: Box plot of *Detection* for the three *Strength* conditions.

conditions and the seven key position conditions. The order of the seven key positions was randomized for each vibration condition, and the order of vibration conditions was also randomized. Each participant performed 945 trials (5 sessions $\times$ 3 tasks $\times$ 9 vibration conditions $\times$ 7 key groups). The experiment took approximately 40 minutes for each participant.

3.6 Results

3.6.1 Vibration Parameters. The trend of *Detection* for different *Strengths* and *Durations* are shown in Figure 5. The overall mean of *Detection* was 79.6%. *Strength* appears to have a major effect on *Detection*, while *Duration* does not. A two-way ANOVA test showed that *Strength* had a statistically significant main effect on *Detection* ($F = 43.870, p < 0.001$), but *Duration* did not ($F = 0.759, ns$). There was no significant interaction effect ($F = 1.549, ns$). Figure 6 shows the distribution of *Detection* for different *Strength* conditions. *Detection* was higher than 90% for all participant when *Strength* was 30% or higher.

3.6.2 Key Position. As the effect of key position on vibration perception may not be seen clearly in the both-hand task, we used the data from the two one-hand tasks only in this analysis. Table 2 shows *Detection* for different *Position* conditions for three

Strength	Position	Top			Top Middle			Middle Bottom			Bottom
		Edge	Center	Avg	Edge	Center	Avg	Edge	Center	Avg	Edge
	10%	67.3	68.9	68.1	62.9	61.3	62.1	54.2	55.8	55	53.6
	30%	98	97.3	97.7	97.1	96.2	96.7	96.2	93.8	95	96.9
50%	99.8	99.6	99.7	99.3	99.1	99.2	99.3	98.7	99	98.4	

Table 2: Detection for the seven Position conditions and the three Strength conditions.

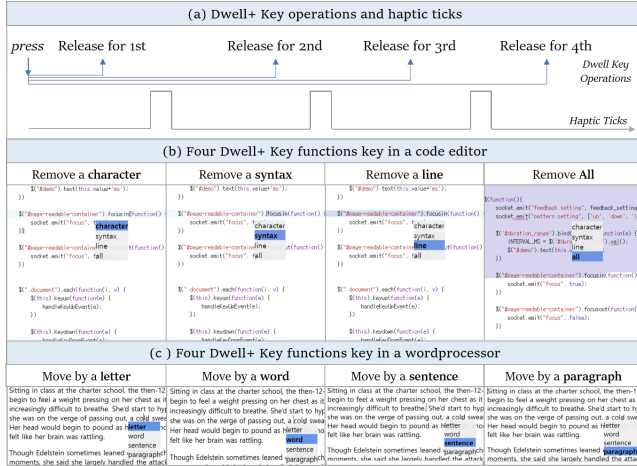


Figure 7: Dwell+ Key technique examples. (a) The Dwell+ Key operations and haptic ticks. When the user presses a key, the current key function changes after each haptic tick until the user releases the key. (b) A Dwell+ Key scenario with the backspace key and (c) another scenario with the arrow keys.

Strength conditions. We aggregated data for different *Duration* levels because *Duration* was shown not to affect *Detection*.

At 10% strength, *Detection* appears to depend on *Position*; it tends to decrease as the key position moves down.

3.7 Conclusion

The *Detection* significantly dropped at the 10% *Strength*. However, users could perceive vibrations for more than 90% of the time when the *Strength* was 30% or higher. In addition, *Duration* did not affect the *Detection*. Therefore, we decided to use a 30% *Strength* (corresponding to 1.17 g) and a duration of 4 ms when designing haptic feedback for HapticPalmrest. We chose 4 ms because a shorter vibration is desired, considering the speed of keyboard interaction and the perceptual disturbance of prolonged vibrotactile stimuli. For the key positions, *Detection* tends to depend on the vertical position of a target key for 10% *Strength*. However, the dependency disappeared, and participants could perceive vibrations well regardless of *Position* for *Strength* 30% or higher. Therefore, we determined to use a uniform vibration strength with a 30% *Strength* in our design for HapticPalmrest.

4 APPLICATION: DWELL+ KEY

Many previous studies presented haptically augmented physical keys and their applications [2, 7, 8, 13]. We expect HapticPalmrest

will be able to enable the same applications. For instance, haptic feedback in response to a key press may indicate a possible typo, thereby reducing typos. A carefully designed haptic response may change the response of a thin keyboard, thereby improving the typing user experience. We are currently developing further applications of HapticPalmrest. As an example of new possible applications, we designed and implemented Dwell+ Key, which was inspired by Dwell+[5]. With HapticPalmRest, a key's alternative characters (functions) may be entered (executed) by a dwelling operation instead of a modifier key.

Holding down a key on the keyboard results in the repetition of the key. For instance, holding down the backspace key results in repeatedly deleting characters. Instead of repeating an action, the backspace may change its mode at every 'tick'; e.g., character-wise, word-wise, line-wise, and so on, as illustrated in Figure 7b. HapticPalmrest provides the tick feedback. If the backspace key is pressed and released before the first tick, it performs the default action: deleting a letter. It performs the second action if pressed and released after the first tick. The same Dwell+ Key scenario may be applied to other keys, such as the space and arrow keys. Figure 7a shows an example scenario with the arrow key.

We had an informal feedback session with four participants from our university. They tried two Dwell+ Key scenarios for the backspace and arrow keys, respectively, as shown in Figure 7. The base strength level and duration of the HapticPalmrest vibration were 30% and 4 ms, respectively. Participants mentioned that the vibration in the base condition was perceivable but could be missed when they were not paying attention to the signal. We changed the vibration parameters to 100% strength and 12 ms duration to test the stronger vibration condition's acceptability. They preferred the latter condition as it was more discrete and better perceived. They mentioned that the haptic tick feedback enabled them to follow the selection change, even if the tick interval was too short to follow visually.

All participants felt the vibration was not distracting. Some mentioned that it felt like the click feedback of the Force Touch trackpad and created an illusion of the palmrest moving down briefly in response to a key press. One mentioned that the palmrest felt like moving down more when the haptic feedback was stronger.

5 DISCUSSION

5.1 Tactons for HapticPalmrest

In each of the Dwell+ Key scenarios, we used identical vibration parameters regardless of the tick level. Using different vibration parameters for different tick levels could have been more effective. For instance, a shorter vibration could be used for the word-level tick, and a more prolonged vibration could be used for the sentence-level tick. In User Study 2, we could determine the minimum vibration parameters as 30% power and 4 ms duration. As a vibration at 100% power level was still acceptable, and the time scale of keyboard events is usually much larger than 4 ms, we think we have ample parameter space for the design of HapticPalmrest tactons.

In addition to the duration and strength of vibration, we can consider more design dimensions for HapticPalmrest tactons. Taptic Engine actuators were used in this study, but a different actuator with a broader bandwidth could be used in the next prototype. For

instance, in a touchscreen click feedback study [4], a piezoelectric actuator could provide more pleasant feedback than a vibration motor. Another large dimension to explore is the spatial dimension of the palmrest. In the simplest case, the left and right vibrotactile plates could be actuated at different times to provide a leftward or rightward signal. In a more advanced case, each plate may contain an array of actuators to provide spatio-temporal tactions.

5.2 Haptic Feedback for Touchpad Interaction

In User Study 1, participants not only used the keyboard but also the touchpad while they performed the three tasks. Therefore, we could also collect data for palm-resting behaviors while they used the touchpad. A quick analysis of the data revealed that the left hand was on the palmrest for 60 to 70% of the time, and the right hand was on the palm 80 to 90% of the time. On the other hand, the contact area was larger for the left hand; the contact area for the right hand was limited to the lower part of the palmrest. We expect that HapticPalmrest could provide haptic feedback in response to touchpad operations; e.g., when the cursor reaches a GUI target.

6 CONCLUSION

We proposed HapticPalmrest, a novel tactile output interface for laptop keyboard interaction. The main conclusion of User Study 1 was that at least one palm was on the palmrest for more than 90% of the interaction time, and the palm-resting area depended on the target key position. The main conclusion of User Study 2 was that a vibration power of 30% (corresponding to 1.17 g) and a vibration duration of 4 ms was sufficient for reliable perception of palmrest vibrations. We finally designed and implemented the Dwell+ Key technique as an HapticPalmrest application example and demonstrated the potential of HapticPalmrest. The empirical findings from the two user studies will provide a solid ground for continued research to make HapticPalmrest a practical component for the laptop, the primary computing platform for productivity applications.

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REFERENCES

- [1] Ahmet Guzerlerler, William R. Provancher, and Cagatay Basdogan. 2016. Perception of Skin Stretch Applied to Palm: Effects of Speed and Displacement. In *Proceedings, Part I, of the 10th International Conference on Haptics: Perception, Devices, Control, and Applications - Volume 9774* (London, UK) (EuroHaptics 2016). Springer-Verlag, Berlin, Heidelberg, 180–189. https://doi.org/10.1007/978-3-319-42321-0_17
- [2] Alexander Hoffmann, Daniel Spelmezan, and Jan Borchers. 2009. TypeRight: A Keyboard with Tactile Error Prevention. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Boston, MA, USA) (CHI '09). Association for Computing Machinery, New York, NY, USA, 2265–2268. <https://doi.org/10.1145/1518701.1519048>
- [3] Sunjun Kim and Geehyuk Lee. 2013. Haptic Feedback Design for a Virtual Button along Force-Displacement Curves. In *Proceedings of the 26th Annual ACM Symposium on User Interface Software and Technology* (St. Andrews, Scotland, United Kingdom) (UIST '13). Association for Computing Machinery, New York, NY, USA, 91–96. <https://doi.org/10.1145/2501988.2502041>
- [4] Emilia Koskinen, Topi Kaaresoja, and Pauli Laitinen. 2008. Feel-Good Touch: Finding the Most Pleasant Tactile Feedback for a Mobile Touch Screen Button. In *Proceedings of the 10th International Conference on Multimodal Interfaces* (Chania, Crete, Greece) (ICMI '08). Association for Computing Machinery, New York, NY, USA, 297–304. <https://doi.org/10.1145/1452392.1452453>
- [5] Yi-Chi Liao, Yen-Chiu Chen, Liwei Chan, and Bing-Yu Chen. 2017. Dwell+: Multi-Level Mode Selection Using Vibrotactile Cues. In *Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology* (Québec City, QC, Canada) (UIST '17). Association for Computing Machinery, New York, NY, USA, 5–16. <https://doi.org/10.1145/3126594.3126627>
- [6] Kouta Minamizawa, Sho Kamuro, Naoki Kawakami, and Susumu Tachi. 2008. A Palm-Worn Haptic Display for Bimanual Operations in Virtual Environments. In *Haptics: Perception, Devices and Scenarios*, Manuel Ferre (Ed.). Springer Berlin Heidelberg, Berlin, Heidelberg, 458–463.
- [7] Daichi Ogawa, Vibol Yem, Taku Hachisu, and Hiroyuki Kajimoto. 2015. Multiple Texture Button by Adding Haptic Vibration and Displacement Sensing to the Physical Button. In *SIGGRAPH Asia 2015 Haptic Media And Contents Design* (Kobe, Japan) (SA '15). Association for Computing Machinery, New York, NY, USA, Article 12, 2 pages. <https://doi.org/10.1145/2818384.2818394>
- [8] Chaeyong Park, Jeongwoo Kim, Dong-Geun Kim, Seungjae Oh, and Seungmoon Choi. 2022. Vibration-Augmented Buttons: Information Transmission Capacity and Application to Interaction Design. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 435, 13 pages. <https://doi.org/10.1145/3491102.3501849>
- [9] Chaeyong Park, Jinhyuk Yoon, Seungjae Oh, and Seungmoon Choi. 2020. Augmenting Physical Buttons with Vibrotactile Feedback for Programmable Feels. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology* (Virtual Event, USA) (UIST '20). Association for Computing Machinery, New York, NY, USA, 924–937. <https://doi.org/10.1145/3379337.3415837>
- [10] Gunhyuk Park, Seungmoon Choi, Kyunghun Hwang, Sunwook Kim, Jaechon Sa, and Moonchae Jung. 2011. Tactile Effect Design and Evaluation for Virtual Buttons on a Mobile Device Touchscreen. In *Proceedings of the 13th International Conference on Human Computer Interaction with Mobile Devices and Services* (Stockholm, Sweden) (MobileHCI '11). Association for Computing Machinery, New York, NY, USA, 11–20. <https://doi.org/10.1145/2037373.2037376>
- [11] Bukun Son and Jaeyoung Park. 2018. Haptic Feedback to the Palm and Fingers for Improved Tactile Perception of Large Objects. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology* (Berlin, Germany) (UIST '18). Association for Computing Machinery, New York, NY, USA, 757–763. <https://doi.org/10.1145/3242587.3242656>
- [12] Ronald T. Verrillo, Anthony J. Fraioli, and Robert L. Smith. 1969. Sensation magnitude of vibrotactile stimuli. *Perception & Psychophysics* 6 (1969), 366–372.
- [13] Chien Min Wu and Shana Smith. 2015. A haptic keypad design with a novel interactive haptic feedback method. *Journal of Engineering Design* 26 (6 2015), 169–186. Issue 4-6. <https://doi.org/10.1080/09544828.2015.1030372>